MLPY Framework Architecture

Core Components
Optional Modules

High-Level API Layer

resample()

benchmark()

Tasks

- TaskClassif
- TaskRegr
- TaskClassifSpatial
- TaskRegrSpatial

Learners

- Sklearn wrappers
- Native implementations
- XGBoost, LightGBM
- H2O, TGPY (optional)

Resamplings

- Standard: CV, Holdout
- Bootstrap
- SpatialKFold
- SpatialBlockCV

Measures

- Classif: Accuracy, AUC
- F1, Precision, Recall
- Regr: RMSE, MAE
- R^2, Bias, MAPE

Spatial ML Focus

AutoML
Tuning, Feature Engineering

Pipelines
PipeOp, GraphLearner

Backends
Dask, Vaex (optional)

Callbacks

Validation

Visualization

Persistence

Typical Workflow

1. Create Task
2. Choose Learner
3. Define Resampling
4. Select Measures
5. Run resample()
6. Analyze Results